Proceedings of the International Conference on Natural Language Processing & Engineering · Published 2021

Grounded Retrieval-Augmented Generation with Strict Attribution and Hallucination Mitigation

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ABSTRACT

Retrieval-augmented language models frequently generate fluent text that incorporates ungrounded prior knowledge rather than retrieved evidence. We present Attrib-RAG, an end-to-end framework enforcing strict citation attribution by conditioning decoder generation on aligned passage spans. Our architecture pairs dense dual-encoder retrieval with token-level source attribution vectors that penalize ungrounded factual assertions. On open-domain scientific QA benchmarks, Attrib-RAG increases sentence-level attribution accuracy from 63% to 94.1% while eliminating unsupported claims. However, strict attribution conditioning causes the model to produce overly conservative responses when retrieved passages contain partial or fragmented evidence.

Keywords— Natural Language Processing, RAG, Retrieval-Augmented Generation, Attribution

INTRODUCTION

While retrieval-augmented generation (RAG) grounds language models in external knowledge, standard cross-attention mechanisms allow generators to drift into hallucinating information from pre-trained weights.

METHODOLOGY

Attrib-RAG enforces an attention attribution mask that penalizes generator hidden states lacking high cosine alignment with retrieved chunk representations. Inline citations are generated as constrained pointer tokens pointing directly to source chunk IDs.

RESULTS

Benchmarked against standard RAG and Fusion-in-Decoder on scholarly QA benchmarks, Attrib-RAG achieved 94.1% attribution accuracy and reduced hallucinated entities by 82%.

LIMITATIONS

When retrieved context is incomplete or ambiguously phrased, the model exhibits hyper-conservatism, frequently generating 'insufficient context' disclosures rather than synthesizing partial logical deductions.

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